

A Packing of 7 Four-Spaces in $\mathbb{R}^{13}$

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October 9, 2026

Abstract

We give 7 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.230536. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.164606, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 7 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.230536 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 7$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.164606, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 7/6 = 3.230769$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N7.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
323053609839238926561007369608448533212574715874650179943733174261541000
069547042728907043866649322696194113069271438979706471736510721620601646
5771709869530930713696465590903468058753163835199
```

denominator:

```
100000000000044830294694032275104071878498968183775351025635314361140033
393766188052347495466096032832633720162188693968472201353081223093706393
9457043760389704603355279207589776330525025387904
```

B Coordinates

7 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N7.txt`.

```
300718984965 259976131252 489662981899 221087530107-352085348624 477396539140 311476041222 179705832594194063853974 -327436085998 17078647832 -578296614175
-425283096101 -71976754916 382889777168 -335044225921905240527 398284594526 -207396885221 -2069529707345899952566 387108826345 -185357369917 -83046800882
-67056567013 -24783531503 421130084701 -70718171161-37552366910 461386463362 25832075363 243494836314 155130825399 122226179863 -628635318773 49566487855
124556440331 156564141075 79634063748 352912188068 -27399194681 44345371609 17051330919 -598293379873 -288147294983 169599599260 -165777269901 -183297742568
36593682571 -243173695207 -253989063548 286645108574177469033347 -197676238085 -165878652348 -3573540855816273985 -164431064415 113564925983 -166458141007
246113160446 8659649642 47046161550 -101160135603 -47462781405 321656945783 -542147164471 -273010784993228896027660 42780828530 13836911924 403957402141
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191461768516 -408535589084 332113080149 -386026513056180477416 -318508165164 -25047272332 2179600463527150419929 -443259152114 -257688572831 213047579653
-102566883652 497185551872 -205036628573 -1221535405330708318903 334848542241 -548844764835 -2235247654535789224490 -144318743392 72594168862 215698383831
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63570939888 217139281865 210964686614 393115707176 -233182572015 -640922377676 -146360799912 165017123237400503888 -169660937580 192856492894 200549276360
227558149388 440684660297 115368161137 -10582571429390133434204 309868638339 -48543475760 -84838926045878206475616 -39863906871 230330774839 -77197277908
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134489569129 345880134446 -237189083732 27721601635529314337661 475726021293 -30507224980 -5605897195640136010930 80233640770 3895740938 293631625142
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